	Player Category	Player Example	Player Roles	Player Roles Details	Following Roles	Player Category that plays the following role
Primary Project Maintainers	Internet Infrastructure	• Network service providers	• [DISCOVER] Identify general infrastructure vulnerabilities	• Detect insecure configurations in DNS, BGP, or other core internet protocols. • Identify exposed or misconfigured infrastructure components across networks. • Log vulnerabilities in shared infrastructure used by many industries, including blockchain operators. • Share general vulnerability intelligence with sector-agnostic CERTs or infrastructure coalitions.	[REMEDIATE]	Internet Infrastructure
			• [DISCOVER] Detect general infrastructure threats	• Detect high-volume traffic anomalies or botnet activity indicative of DDoS attacks. Monitor for route hijacks, DNS spoofing, or censorship patterns. • Identify mass-scanning or exploitation activity affecting public-facing infrastructure. • Share threat intelligence with cross-industry partners, including blockchain-focused entities.	[REPORT] Share observed network-level threats with relevant stakeholders	Internet Infrastructure
			• [REMEDIATE] Address infrastructure vulnerabilities and respond to detected threats	• Apply patches or configuration changes to fix detected infrastructure vulnerabilities (e.g., BGP misconfigurations, exposed DNS services). • Respond to active threats such as DDoS attacks, traffic hijacking, or network abuse, including blackholing, rate-limiting, or rerouting traffic. • Isolate compromised systems or services used in broader attacks affecting multiple sectors. • Coordinate with incident response teams and upstream providers to contain and mitigate cascading threats. • Notify affected downstream customers when remediation actions may impact service availability.	[REPORT]	Internet Infrastructure Security Expert
			• [REPORT] Share identified infrastructure vulnerabilities with relevant stakeholders	• Report discovered vulnerabilities (e.g., BGP leaks, DNSSEC failures) to shared infrastructure operators and affected sectors. • Notify CERTs, ISACs, and backbone providers of systemic risks. • Contribute vulnerability intelligence to multi-sector databases or coordinated disclosure platforms.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC

					[ASSESS]	Blockchain Protocol Blockchain Network
						Security Expert
						Certification Body
			• [REPORT] Share observed network-level threats with relevant stakeholders	• Share threat intelligence on traffic anomalies, attack sources, or known malicious infrastructure. • Report incidents of internet-layer abuse (e.g., censorship, hijacks) that may disrupt blockchain node or wallet access. • Inform infrastructure-reliant blockchain actors (e.g., exchanges, bridges, RPC services) about threats that may degrade service or expose users. • Escalate emerging threats to cross-sector CERTs or coordination hubs (e.g., Crypto-focused ISAC) for broader dissemination.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
					[ASSESS]	Blockchain Protocol Blockchain Network
						Security Expert
			• [ASSESS] Assess risks based on shared vulnerabilities and threats intelligence	• Analyze disclosed vulnerabilities (e.g., DNS flaws, routing misconfigurations) that may affect service availability for blockchain participants. • Review threat reports (e.g., botnet campaigns, targeted DDoS activity) for potential impact on core infrastructure used by blockchain networks. • Coordinate internally to determine whether shared blockchain-related incidents signal broader systemic risks. • Update risk models and incident response readiness to reflect newly shared threat intelligence that could propagate across dependent sectors. • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.	[DISCOVER]	Internet Infrastructure
					[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm
						Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
					[END]	-
					[CONTRIBUTE]	Internet Infrastructure

				• [CONTRIBUTE] Develop and maintain general internet security and reliability standards	• Participate in technical standards bodies (e.g., IETF, ICANN, MANRS) to define best practices for routing, DNS, and peering security. • Contribute operational insights to the development of mitigation standards for DDoS, BGP hijacking, and network abuse. • Support efforts to standardize infrastructure-layer threat sharing and response protocols. • Promote adoption of secure configurations and incident handling frameworks across sectors. • Ensure that general infrastructure standards are transparent and interoperable for any dependent industries, including blockchain.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Apply general internet infrastructure standards for secure and resilient network operations	• Implement routing security best practices, such as RPKI and BGP filtering, in line with initiatives like MANRS. • Use standardized DNS security protocols, including DNSSEC, to prevent hijacking and spoofing. • Apply incident response and abuse handling procedures based on NIST, ISO, or sector-neutral internet standards. • Maintain infrastructure availability and integrity by aligning operations with protocols defined by IETF, ICANN, or regional bodies. • Support interoperable communications and threat mitigation, enabling coordination across sectors, including blockchain applications.	[END]	-
	Blockchain Protocol	**Blockchain**	• L1 Blockchain foundation (Ethereum Foundation, Solana Foundation, etc) • L2 Blockchain foundation (Arbitrum Foundation, etc) • Node Providers (incl RPC/Archive Nodes) • Other Open-Source Service Providers (node software, etc)	• [DISCOVER] Undercover vulnerabilities in consensus, client code, or infrastructure	• Conduct internal audits and security reviews of core protocol code, consensus mechanisms, and node implementation. • Analyze the behavior of transaction ordering, mempool handling, and validator logic to detect design-level flaws. • Monitor community bug reports, bounty programs, and code repositories for disclosed or emerging vulnerabilities. • Engage in fuzzing, formal verification, and symbolic execution to proactively identify issues in state transitions and smart contract platforms.	[REMEDIATE]	Blockchain Protocol (Blockchain)
				• [DISCOVER] Detect network anomalies or protocol-level threat	• Detect abnormal validator behavior, spam attacks, gas congestion manipulation, or potential MEV exploitation. • Monitor fork attempts, Sybil threats, cross-layer reorgs, and malicious node behavior. • Identify attack patterns against consensus mechanisms (e.g., long-range attacks, proposer censorship). • Track infrastructure abuse, e.g., RPC endpoint overuse, archive node manipulation, or misbehaving relayers.	[REPORT] Disseminate threat intelligence on consensus or infrastructure attacks	Blockchain Protocol (Blockchain)
				• [REMEDIATE] Develop and release protocol upgrades	• Design, implement, and test protocol upgrades (e.g., hard forks, soft forks, client patches) to address identified vulnerabilities or design flaws. • Coordinate upgrade releases across multiple client implementations and node operator communities. • Execute emergency upgrades when systemic bugs or economic attacks are discovered. • Conduct regression testing, consensus simulations, and backwards compatibility checks before rollout. • Communicate upgrade timelines and change logs to the entire ecosystem, including exchanges, L2s, and app developers.	[REPORT]	Blockchain Protocol (Blockchain) Security Expert
				• [REPORT] Share protocol or node vulnerabilities to stakeholders	• Disclose confirmed protocol or implementation vulnerabilities to node operators or affected stakeholders through responsible disclosure. • Publish detailed security advisories for protocol bugs and provide upgrade guidance or emergency patches. • Share vulnerability findings with ecosystem stakeholders (e.g., bridges, rollups, apps) if cross-protocol implications exist. • Collaborate with upstream and downstream dependencies to coordinate disclosure timelines.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network

							Blockchain App Consumers/Relying Party Security Expert Auditor Insurance Company Certification Body
				• [REPORT] Disseminate threat intelligence on consensus or infrastructure attacks	• Disseminate information about active threats (e.g., validator misbehavior, slashing incidents, spam attacks) to the community and infrastructure providers. • Issue alerts about zero-day threats, consensus risks, or exploits affecting cross-chain or on-chain operations. • Coordinate with exchanges, analytics firms, L2 teams, and custodians to relay time-sensitive intelligence. • Report security incident summaries and mitigations in public postmortems or emergency forums (e.g., Discord, GitHub, Twitter).	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
				• [ASSESS] Assess risks using shared threat and vulnerability data	• Evaluate the impact of reported threats (e.g., DDoS, censorship, MEV abuse, chain instability) on protocol performance and user safety. • Cross-reference shared threat indicators (e.g., malicious contract addresses, slashing events) with on-chain logs and telemetry data. • Update risk dashboards, node configurations, and response protocols in reaction to shared incidents or coordinated threat alerts. • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.	[DISCOVER]	Blockchain Protocol (Blockchain)
						[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team

							Crypto-focused ISAC
						[END]	-
						[CONTRIBUTE]	Blockchain Protocol (Blockchain)
				• [CONTRIBUTE] Contribute protocol-level insights to global standardization efforts	• Propose and co-author formal protocol improvement proposals. • Participate in working groups to define consensus, fee markets, cryptography, interoperability, or governance standards. • Align protocol behaviors with broader technical or regulatory frameworks. • Collaborate with standards bodies, other app/protocol developers, and infrastructure providers to co-design scalable, secure architecture.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Implement and uphold security and protocol standards	• Implement protocol and node changes according to approved technical standards. • Align node software, RPC interfaces, and block production logic with community-agreed specifications. • Maintain backward and forward compatibility with ecosystem-wide tooling and protocol versions. • Ensure security and operational processes conform to best practices defined in sectoral security guidelines.	[END]	-
		DeFi	• DeFi Developers • Stablecoin Issuers	• [DISCOVER] Identify security gaps in DeFi logic, contracts, or integrations	• Conduct internal audits and simulations of smart contracts, governance logic, and upgradeable proxy patterns to detect vulnerabilities. • Monitor economic behavior (e.g., oracle manipulation, governance capture risks, liquidity fragmentation) for protocol design flaws. • Run bug bounty programs and integrate community-discovered issues for proactive vulnerability discovery.	[REMEDIATE]	Blockchain Protocol (DeFi)
				• [DISCOVER] Identify exploit patterns and attack vectors in DeFi systems	• Monitor abnormal user behavior, flash loan activity, and transaction patterns to detect potential exploits. • Track malicious governance proposals, MEV-based attacks, and oracle failures affecting collateral or pricing. • Detect smart contract abuse or protocol integrations under attack.	[REPORT] Report detected threats to stakeholders across the DeFi ecosystem	Blockchain Protocol (DeFi)
				• [REMEDIATE] Deploy security fixes and protocol upgrades	• Implement emergency upgrades or parameter changes (e.g., pausing modules, adjusting caps, disabling features) in response to exploits. • Execute governance processes or use admin privileges (if retained) to deploy security fixes and mitigation patches. • Publish upgrade documentation, test suites, and migration plans to ensure downstream stability. • Collaborate with multisig holders, governance participants, and devops teams to coordinate safe rollout of changes.	[REPORT]	Blockchain Protocol (DeFi)
							Security Expert
				• [REPORT] Report security flaws in DeFi protocols to stakeholders	• Report discovered vulnerabilities in shared dependencies (e.g., oracles, router contracts) to maintainers or upstream projects. • Notify liquidity providers, governance participants, and integrators of design flaws or bugs before public disclosure. • Publish technical postmortems and remediation notes after public fixes.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App

							Consumers/Relying Party Security Expert Auditor Insurance Company
							Certification Body
				• [REPORT] Report detected threats to stakeholders across the DeFi ecosystem	• Alert peer protocols, integrators, and RPC providers about observed attack campaigns or exploit attempts. • Report scam contracts, malicious tokens, or forked imposters using phishing or impersonation tactics. • Coordinate across ecosystem stakeholders (e.g., bridges, L2s) when threats span multiple chains.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
				• [ASSESS] Monitor DeFi vulnerabilities informed by ecosystem disclosures	• Review threat reports and vulnerability disclosures from other DeFi protocols, auditors, or analytics firms to assess potential impact on their own systems. • Cross-reference shared attack vectors (e.g., oracle manipulation, governance exploits, cross-chain vulnerabilities) with internal architecture and contract logic. • Monitor ecosystem-wide alerts (e.g., from ISACs, bug bounty disclosures, or Twitter/Discord) to detect similar exposure risks. • Analyze disclosed incidents to determine whether emergency response or protocol parameter adjustments (e.g., collateral ratios, caps) are necessary. • Collaborate with ecosystem partners and integrated platforms (e.g., aggregators, bridges, stablecoins) to evaluate cascading risk from shared vulnerabilities. • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.	[DISCOVER]	Blockchain Protocol (DeFi)
						[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm
							Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC

						[END]	-
						[CONTRIBUTE]	Blockchain Protocol (DeFi)
				• [CONTRIBUTE] Collaborate on open standards for DeFi security and functionality	• Propose new patterns for smart contract safety, liquidity risk management, collateral standards, and oracle usage in forums. • Co-develop secure proxy upgrade frameworks or governance modules for modular reuse. • Share security best practices such as token interface extensions, and L2 integration guides as open standards.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Adopt recognized standards for DeFi contracts and integrations	• Implement controls and development processes aligned with standards. • Follow formal verification, auditing, and release procedures recommended by ecosystem governance or regulatory guidance. • Ensure off-chain integrations (e.g., pricing, governance UIs, routers) adhere to secure infrastructure and deployment standards.	[END]	-
	Blockchain Network	Network Operator	• Miners • Validators (Proposers / Builders) • Mining/Validation Pool Operators • Sequencers / Data Managers • Relayers	• [DISCOVER] Identify weaknesses in network infrastructure or protocol behavior	• Monitor block production and transaction behavior to detect inconsistencies or anomalies that may reveal protocol or client vulnerabilities. • Observe validator or peer node misbehavior (e.g., equivocation, missed attestations) that could indicate flaws in consensus logic or slashing conditions. • Identify abnormal patterns in mempool activity such as malformed transactions or invalid blocks submitted to the network. • Detect vulnerabilities in relaying or data availability mechanisms (e.g., incorrect proofs, latency-based manipulation, missing commitments). • Flag infrastructure-level issues like RPC failures, syncing problems, or signature validation errors that point to deeper client or network bugs.	[REMEDIATE]	Blockchain Network (Network Operator)
				• [DISCOVER] Detect network-level threats and suspicious activity	• Monitor for malicious validator or miner activity, such as proposer censorship, time manipulation, or double-signing. • Detect suspicious transaction patterns in the mempool or blocks (e.g., frontrunning, spam, phishing contract deployment). • Identify attempts to exploit MEV, bypass consensus rules, or manipulate block content and inclusion order. • Observe anomalies in relayer behavior (e.g., delayed or dropped cross-chain messages) that may indicate targeted attacks. • Flag data availability or sequencing issues that could signal emerging threats.	[REPORT] Report observed threats to the network or ecosystem	Blockchain Network (Network Operator)
				• [REMEDIATE] Take immediate action to apply security fixes and mitigate threats	• Upgrade client software promptly to apply emergency patches that fix critical vulnerabilities. • Temporarily pause or adjust operations (e.g., block production or relaying) to prevent further risk propagation. • Follow coordinated response instructions issued by protocol developers or validator communities during live threats. • Halt validation or relaying of harmful transactions when exploit patterns or protocol violations are detected. • Apply real-time mitigations such as filtering mempool activity, adjusting fees, or changing block template settings. • Ensure continued uptime and honest participation to maintain network stability and prevent consensus breakdown.	[REPORT]	Blockchain Network (Network Operator) Security Expert
				• [REPORT] Report observed vulnerabilities in network infrastructure or protocol behavior	• Report bugs or faults in client software (e.g., consensus inconsistencies, syncing issues, block validation errors) to core developers or client teams. • Alert infrastructure providers or partners (e.g., L2 teams, relayers, bridges) about vulnerabilities in data availability, relaying, or message handling. • Coordinate responsibly to avoid disclosing exploitable information prematurely while ensuring that critical actors are informed.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol

							Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company
							Certification Body
				• [REPORT] Report observed threats to the network or ecosystem	• Alert core developers and protocol teams to signs of validator misbehavior, censorship, or attack attempts. • Report observed MEV manipulation, transaction flooding, or reordering that could impact fairness or consensus safety. • Share intelligence with peer operators on detected phishing contracts, attack patterns, or chain reorgs. • Notify relayer or bridge teams of message tampering, delays, or unusual routing behavior.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
							Certification Body
				• [ASSESS] Evaluate risks to the network using shared threat and vulnerability intelligence	• Review disclosed vulnerabilities or threat reports from protocol teams, peers, or security alliances to assess potential impact. • Monitor node behavior and block content in light of known threats, such as spam attacks, invalid transactions, or MEV abuse. • Compare shared attack indicators (e.g., suspicious wallet addresses or contract hashes) with local node activity and mempool data. • Adjust validator or relayer settings in response to known risks (e.g., temporarily disabling a vulnerable feature). • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.	[DISCOVER]	Blockchain Network (Network Operator)
						[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm
							Academia Law Enforcement Authority

							Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
						[END]	-
						[CONTRIBUTE]	Blockchain Network (Network Operator)
				• [CONTRIBUTE] Help shape operational and security standards through network participation	• Provide feedback on proposed protocol or consensus changes, sharing insights based on real-world validator or miner behavior. • Participate in testnets and standardization efforts (e.g., Ethereum's AllCoreDevs, rollup standard discussions, IETF/CCSS initiatives). • Support development of MEV-related or data availability standards by reporting what works operationally and what doesn't. • Share lessons learned from edge cases or live incidents to inform slashing conditions, fork choice rules, or fee market standards. • Help validate the feasibility and safety of upgrades and new designs by testing draft implementations and reporting outcomes.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Operate in alignment with established protocol and security standards	• Participate in on-chain or off-chain governance to approve or reject proposed protocol upgrades with security implications • Run governance clients or participate in snapshot voting using staked tokens or delegated voting rights (e.g., in protocols like Cosmos, Polkadot, Optimism). • Review and evaluate upgrade proposals from core developers, security councils, or DAOs to assess potential risks or unintended consequences. • Monitor testnet deployments and participate in audits or test-phase voting to validate security upgrades before mainnet activation.	[END]	-
		Data Provider	• Blockchain Explorers • Indexers	• [DISCOVER] Identify anomalies and emerging threats from on-chain data	• Operate monitoring systems to detect node downtime, block production irregularities, reorgs, and peer misbehavior. • Track network latency, transaction inclusion delays, or block proposer failures across clients and geographies. • Detect signs of censorship, MEV abuse, and frontrunning through analysis of mempool activity and block content. • Monitor for validator collusion, stake centralization risks, or attacks on consensus (e.g., long-range, eclipse, or equivocation). • For indexers: detect data corruption, missed updates, or manipulation in query responses and indexing logic.	[REPORT]	Blockchain Network (Data Provider)
				• [REPORT] Share structured threat intelligence and transparency data	• Share real-time alerts and postmortems with client teams, foundations, and security alliances when threats are observed. • Publish reports on observed MEV extraction patterns, exploit attempts, or systemic risks affecting protocol security or decentralization. • Notify core dev teams or protocol governance of anomalies discovered via infrastructure monitoring tools. • Coordinate with peer operators and open-source communities during live attacks or technical crises (e.g., “war rooms”).	[DISCLOSE]	Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party

							Security Expert Auditor Insurance Company Blockchain Analytics Firm
				• [CONTRIBUTE] Help standardize/structure blockchain data for use across the ecosystem	• Help define schema standards for threat reporting, risk scoring, and address attribution (e.g., wallet clustering, event labeling). • Support or co-develop open data protocols that enable interoperability between wallets, exchanges, and regulators. • Contribute to transparency standards (e.g., verified contracts, audit flags) used in explorers and monitoring tools.	[STANDARDIZE]	International Standards Organization National Standards Organization
	Blockchain App	Self-Custodial	• Self-Custodial Wallet Providers • Transaction Managers (ex. Safe, Ginco) • Other Hardware Providers	• [DISCOVER] Detect security flaws in wallet software, hardware, or transaction infrastructure	• Audit wallet code and transaction workflows to detect bugs in key management, signing logic, or gas calculation. • Identify UI vulnerabilities that may enable phishing or transaction spoofing (e.g., misleading prompts, approval trickery). • Analyze firmware and hardware components to find flaws in entropy generation, memory protection, or physical access resistance. • Detect misuse of integrated libraries or SDKs, especially where reused modules (e.g., for account abstraction or MPC) introduce attack surfaces. • Monitor bug bounty reports and user feedback for indicators of previously unknown wallet-layer vulnerabilities.	[REMEDIATE]	Blockchain App (Self-custodial)
				• [DISCOVER] Detect active threats targeting wallet users, interfaces, or infrastructure	• Monitor for phishing websites and fake apps impersonating the wallet or transaction manager interfaces. • Detect scam contracts and malicious transaction patterns that attempt to exploit users through wallet prompts. • Identify malware and browser extensions that attempt to intercept seed phrases, alter transaction details, or inject code. • Track emerging attack techniques, such as drainer scripts using stealth approvals, token re-issuance tricks, or social engineering. • Analyze telemetry or user-reported incidents to uncover coordinated threat campaigns or interface manipulation. • Watch for firmware tampering attempts or attacks on hardware wallet supply chains and bootloader vulnerabilities.	[REPORT]	Blockchain App (Self-custodial)
				• [REMEDIATE] Patch security flaws and strengthen user protection mechanisms	• Deploy software or firmware updates that patch vulnerabilities in wallet logic, signing flows, recovery mechanisms, or device firmware. • Improve user-facing protections such as transaction simulations, scam warnings, address labeling, and approval risk alerts. • Enhance logic for detecting suspicious behaviors (e.g., repeated approvals, risky contract interactions, or account draining patterns). • Integrate optional wallet features that support AML-aligned behavior, such as wallet screening tools or optional integration with third-party compliance services. • Refactor transaction manager infrastructure to reduce risk of front-end spoofing, key misuse, or replay attacks. • Collaborate with open-source communities to upstream patches and strengthen shared libraries used across wallet ecosystems.	[REPORT]	Blockchain App (Self-custodial) Security Expert
				• [REPORT] Report discovered security vulnerabilities to relevant stakeholders	• Coordinate responsible disclosure of vulnerabilities to protocol developers, hardware component suppliers, or library maintainers. • Notify affected ecosystem partners (e.g., integrated dApps, bridges, SDK providers) when a discovered issue may impact shared users. • Submit reports to public vulnerability databases or CVE programs when flaws affect open-source infrastructure. • Share anonymized vulnerability summaries with threat intelligence groups, ISACs, or incident coordination channels. • Publish security advisories and user guidance when wallet-side patches are released (e.g., firmware or software updates).	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol

					• Work with bug bounty platforms or white-hat researchers to ensure discovered vulnerabilities are verified and disclosed ethically.		Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company
							Certification Body
				• [REPORT] Share detected threats targeting wallet users or infrastructure	• Report phishing sites, scam apps, or fake wallet interfaces to domain takedown services, browser providers, and security alliances. • Alert other ecosystem actors (e.g., analytics firms, dApps, bridges) about known malicious contracts, drainer patterns, or exploitable dApp interactions. • Share intelligence with threat monitoring groups, such as Crypto-ISACs, SEAL Alliance, or community-based rapid response channels. • Publish security bulletins warning users and integrators of ongoing attack campaigns or high-risk behaviors. • Contribute to blacklists and warning systems by flagging malicious addresses, approval traps, or signature spoofing attempts. • Coordinate with hardware/software vendors if detected threats involve third-party firmware, browser extensions, or supply chain vectors.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
				• [ASSESS] Evaluate security exposure based on disclosed vulnerabilities and threat intelligence	• Review vulnerability disclosures and threat reports from ecosystem actors to assess potential impact. • Compare shared attack patterns (e.g., phishing flows, drainer contracts, approval exploits) against existing wallet flows and UI logic. • Evaluate dependencies (e.g., SDKs, transaction routers, MPC libraries) for exposure to reported vulnerabilities. • Conduct internal audits or simulations in response to shared incidents to verify whether users or systems are at risk. • Adjust threat models and user prompts based on newly understood attacker behaviors or risks circulating in the ecosystem. • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.	[DISCOVER]	Blockchain App (Self-custodial)
						[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm
							Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit

							Gov. Lab/Response Team Crypto-focused ISAC
						[END]	-
						[CONTRIBUE]	Blockchain App (Self-custodial)
				• [CONTRIBUTE] Help define standards for secure wallet operations and user protection	• Participate in working groups and open forums to shape wallet-related standards. • Help define hardware wallet and transaction manager security benchmarks, such as seed storage, device integrity, and tamper resistance. • Collaborate with protocols, dApps, and ecosystem partners to create interoperable and modular standards for secure wallet integration.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Implement recognized security and interoperability standards	• Implement widely accepted wallet standards. • Comply with hardware and firmware security frameworks for secure device communication. • Adopt transaction safety guidelines (e.g., approval simulations, phishing warnings). • Ensure secure interoperability with DeFi protocols and bridges through standardized wallet connection interfaces. • Regularly update implementations to match improvements in standards or new requirements introduced by protocols or governance groups.	[END]	-
		DApp Developers	• DApp Frontend Developers	• [DISCOVER] Detect vulnerabilities in smart contract interactions and frontend components	• Audit frontend code and wallet connection logic to identify security flaws (e.g., transaction injection, spoofed prompts, unsafe signing behavior). • Review smart contract interactions to catch dangerous defaults (e.g., unverified contract calls, unsafe approvals). • Test for manipulation or bypass vulnerabilities in authentication flows, session handling, or off-chain signing processes. • Detect inconsistencies between frontend display and contract behavior that could mislead users or hide malicious actions. • Monitor user feedback and bug reports to uncover interface-level vulnerabilities or Web3 library misuses. • Assess the use of third-party SDKs, APIs, or middleware for potential vulnerabilities in wallet integration or data display.	[REMEDIATE]	Blockchain App (DApp developers)
				• [DISCOVER] Detect active threats targeting dApp users or interfaces	• Monitor for phishing sites or spoofed dApp frontends impersonating the legitimate application interface. • Detect malicious smart contract interactions triggered through the dApp (e.g., drainer contracts, token impersonation). • Identify injection or manipulation attempts within wallet connection flows or transaction builders. • Observe abnormal transaction patterns initiated via the frontend that could indicate bot activity or attack scripts. • Track external threats targeting integrated services, such as oracles, bridges, or APIs used by the frontend. • Analyze user-reported incidents or behavioral anomalies that suggest targeted threats or evolving scams.	[REPORT]	Blockchain App (DApp developers)
				• [REMEDIATE] Patch security flaw and enhance user protection against abuse	• Patch frontend code vulnerabilities related to transaction manipulation, UI spoofing, and unsafe wallet interactions. • Refactor smart contract interfaces and user flows to minimize risks from unsafe approvals, hidden function calls, or malicious input. • Implement transaction safety features, like warnings for risky contracts, simulation previews, or trusted contract verification.	[REPORT]	Blockchain App (DApp developers)

					• Integrate blocklist services or KYT (Know Your Transaction) tools to flag interactions with sanctioned or suspicious addresses. • Improve on-chain and off-chain logging to enable transparency and facilitate downstream forensic or compliance analysis.		Security Expert
				• [REPORT] Share vulnerabilities in dApp interfaces or integrations to the ecosystem	• Disclose frontend vulnerabilities (e.g., signing spoofing, injection risks, misconfigured wallets) to internal teams and security researchers. • Report issues in third-party SDKs or APIs (e.g., transaction routers, wallet connectors) to open-source maintainers or vendors. • Coordinate responsible disclosure of smart contract interaction bugs or logic flaws to affected protocol teams. • Submit vulnerability reports to bug bounty platforms, security alliances (e.g., SEAL), or ecosystem forums. • Notify integrators or partners if flaws in the dApp interface could propagate risk across multiple applications. • Publish post-mortems or advisory notes after remediation, helping others learn and avoid similar mistakes.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company
				• [REPORT] Share observed threats impacting dApp users or infrastructure	• Alert security communities and protocol teams about malicious contract interactions, phishing domains, or drainer behavior seen through the dApp. • Report threats related to integrated services (e.g., bridges, oracles, wallet connectors) to relevant stakeholders. • Share real-time threat intelligence with partners, forums, or alliances (e.g., Crypto-ISACs, SEAL, Discord channels). • Publish security bulletins or notices in the dApp UI or official channels to warn users of ongoing threats. • Contribute to threat databases or blacklists by flagging risky addresses, fake tokens, or frontends impersonating the dApp. • Coordinate with browser wallets and infrastructure providers to escalate urgent threats affecting large user groups. • Educate users with in-app prompts and guides that explain security risks (e.g., never sign unknown messages, verify contracts).	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
				• [ASSESS] Evaluate dApp security exposure using shared threat and vulnerability intelligence	• Review disclosed vulnerabilities and incident reports from ecosystem partners to assess potential impact on dApp operations. • Cross-reference attack indicators or malicious contract patterns against the dApp's integrated contracts and external dependencies. • Evaluate the risk of third-party libraries or SDKs used in wallet connection, contract routing, or token display logic. • Monitor ecosystem-wide phishing or drainer campaigns to determine whether the dApp frontend could be spoofed or misused. • Simulate attack scenarios or user interaction edge cases to test the robustness of the interface under known threat conditions.	[DISCOVER]	Blockchain App (DApp developers)
						[ANALYZE]	Consumers/Relying Party Security Expert Auditor

					• Use findings to inform internal security reviews and prioritize patching, interface adjustments, or communication with users. • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.		Blockchain Analytics Firm
							Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
						[END]	-
						[CONTRIBUTE]	Blockchain App (DApp developers)
				• [CONTRIBUTE] Help define secure and interoperable dApp design standards	• Propose improvements to frontend interaction standards, such as safe signing flows, permission displays, or multisig UX. • Contribute design feedback on how users engage with smart contracts, including gas estimations, transaction confirmation, and risk warnings. • Support standardization of reusable components, like wallet adapters, token approval interfaces, and cross-chain dApp connection patterns. • Share incident response lessons to inform community-wide improvements to interface and integration practices.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Implement established standards for secure and interoperable dApp development	• Implement interface and interaction standards like token interaction best practices. • Adopt wallet connection frameworks and permissions systems to ensure compatibility and transparency. • Apply security guidelines for dApp frontends, including approval prompts, transaction previews, and malicious contract detection. • Ensure compatibility with on-chain governance, DeFi protocols, or multisig tools by aligning with standard component libraries and smart contract interfaces. • Continuously update the app to reflect new or evolving standards adopted by the ecosystem.	[END]	
		Cutodial, VASPs	• Centralized Exchanges • Custodial Wallet Providers • Other Custodial Service Providers • Crypto Payment Processors	• [DISCOVER] Detect vulnerabilities in custodial infrastructure and service architecture	• Conduct regular security audits of backend systems, wallet management platforms, KYC/AML modules, and transaction engines. • Identify weaknesses in hot/cold wallet segregation, signing workflows, or MPC/HSM configurations. • Test internal systems for exploitable flaws in access controls, user account management, and withdrawal validation logic. • Review API endpoints and third-party integrations for injection, spoofing, or abuse vectors. • Monitor operational logs and incident patterns for signs of latent vulnerabilities (e.g., unhandled errors, resource exhaustion, misrouted transactions). • Use threat modeling and red-teaming to simulate attacks on asset custody pipelines and user authentication flows.	[REMEDIATE]	Blockchain App (Custodial)

				• [DISCOVER] Detect active threats targeting custodial systems	• Monitor for phishing attacks targeting user login portals, mobile apps, or email communications. • Detect suspicious user activity such as repeated failed logins, anomalous withdrawals, or abnormal transaction behavior. • Identify external abuse attempts, including bot attacks, DDoS, account takeovers, or API abuse. • Track evolving tactics such as new wallet drainer scripts, fake token listings, or social engineering campaigns. • Analyze blockchain transactions in real time to spot interactions with known malicious addresses, mixers, or exploit-linked wallets. • Integrate threat intelligence feeds from analytics firms, law enforcement, or ISACs to stay ahead of coordinated threat actors.	[REPORT] Share threat intelligence related to attacks, fraud, and suspicious behavior	Blockchain App (Custodial)
				• [DISCOVER] Detect vulnerabilities and threats linked to third-party dependencies	• Assess third-party integrations for misconfigurations or exploitable flaws. • Track threat intelligence feeds and advisories related to partners, vendors, and supply chain components. • Perform third-party risk scans or audits to proactively identify insecure endpoints, outdated dependencies, or exposed credentials. • Simulate attack paths involving external services (e.g., phishing via support tools, API spoofing, or oracle manipulation) to validate real-world risks. • Collaborate with vendors to verify security postures, remediate vulnerabilities, and escalate urgent threats if needed.	[ALERT]	Blockchain App (Custodial)
						[REMEDIATE]	Blockchain Protocol Blockchain Network Blockchain App
				• [ALERT] Notify core protocol or project maintainers of identified vulnerabilities or threats	• Report vulnerabilities or threat indicators found in protocols or apps directly to the relevant maintainers or foundations. • Provide detailed technical information (e.g., attack vectors, logs, replication steps) to aid investigation and remediation. • Engage through official disclosure channels, such as GitHub issues, security email contacts, or bug bounty platforms. • Flag coordinated threat campaigns that may impact multiple projects (e.g., drainer contracts, cross-chain attacks). • Follow up to support remediation, offer testing, or coordinate patch timelines as needed.	[REMEDIATE]	Blockchain Protocol Blockchain Network Blockchain App
						[ESCALATE]	Blockchain App (Custodial)
				• [REMEDIATE] Patch security flaws and strengthen AML/CFT controls	• Apply software and infrastructure patches to fix vulnerabilities in wallet architecture, API endpoints, and authentication layers. • Update internal controls and transaction monitoring systems to reflect new threat patterns or regulatory expectations. • Implement secure key management improvements, such as HSM upgrades, MPC enhancements, or improved access restrictions. • Strengthen AML/CFT frameworks by improving real-time transaction screening, risk scoring, and sanctions enforcement. • Revise user onboarding and withdrawal procedures to close gaps exploited by fraudsters or illicit actors. • Collaborate with legal, compliance, and security teams to ensure coordinated remediation across technical and regulatory domains.	[REPORT]	Blockchain App (Custodial)
							Security Expert
				• [ESCALATE] Escalate unresolved third-party vulnerabilities to relevant stakeholders	• Formally document and track unresolved vulnerability disclosures made to third-party vendors or protocol teams. • Notify affected ecosystem stakeholders (e.g., users, partners, infrastructure providers) of the outstanding risk. • Escalate the issue through trusted disclosure networks, such as CERTs, ISACs, or security alliances. • Engage with regulators or supervisory bodies when unremediated vulnerabilities pose systemic or user protection risks. • Share limited public advisories—when appropriate—to alert developers, auditors, and integrators while minimizing exploitation risk. • Update internal vendor risk profiles and adjust dependencies where remediation is unlikely or delayed.	[WARN]	Gov. Lab/Response Team Crypto-focused ISAC

				• [REPORT] Report discovered vulnerabilities in custodial systems to relevant stakeholders	• Report critical vulnerabilities (e.g., flaws in key management, transaction processing, or account security) to internal security and compliance teams. • Coordinate responsible disclosures with third-party service providers, software vendors, or open-source maintainers if the issue lies in shared infrastructure. • Share relevant findings with industry alliances, CERTs, or ISACs when broader systemic risk is possible. • Notify regulators or supervisory bodies when legally required, especially for vulnerabilities that may have affected customer data or funds. • Collaborate with bug bounty platforms or external researchers to validate and document discovered issues. • Publish post-remediation advisories, where appropriate, to inform users and industry peers about resolved risks and best practices.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company
				• [REPORT] Share threat intelligence related to attacks, fraud, and suspicious behavior	• Report observed threats such as phishing domains, account takeovers, and API abuse to peers and external investigators. • Share indicators of compromise (IOCs)—e.g., suspicious wallet addresses, IPs, transaction patterns—with threat intelligence networks or industry ISACs. • Alert integrated service providers or infrastructure partners if they may be affected by observed threats (e.g., malware targeting wallet users). • Notify law enforcement or regulatory bodies of high-risk events like hacks, coordinated laundering attempts, or fraud rings. • Participate in cross-industry threat sharing efforts, helping build blocklists, scam databases, and early warning systems. • Contribute anonymized threat data to monitoring platforms and forensic tools to support broader ecosystem resilience.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
						[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
				• [ASSESS] Evaluate internal exposure in response to external threat and vulnerability intelligence	• Analyze threat intelligence reports from industry sources to determine relevance to internal infrastructure or services. • Assess whether disclosed vulnerabilities in shared libraries, wallet infrastructure, or protocols could affect internal components or integrations. • Cross-check internal logs and telemetry data against known attack signatures, blacklisted addresses, or behavioral indicators. • Evaluate the exposure of user data, key management systems, and transaction workflows based on current threat landscape. • Conduct internal red-teaming or simulations to validate defensive readiness against emerging attack vectors. • Use findings to update risk registers, adjust security controls, and escalate to remediation or incident response where needed. • Revise internal blacklists, whitelists, and due diligence procedures in response to shared threat intelligence.	[DISCOVER]	Blockchain App (Custodial)
						[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm

							Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
						[END]	
						[CONTRIBUTE]	Blockchain App (Custodial)
				• [ASSESS] Monitor third-party exposure using shared vulnerability and threat intelligence	• Evaluate the risk impact of vulnerabilities disclosed in third-party services. • Track threat intelligence related to key partners, including wallet SDKs, blockchain APIs, and liquidity providers. • Monitor integration points (e.g., bridges, payment gateways, oracles) for signs of compromise or known threats. • Cross-reference third-party dependencies with alerts from ISACs, CVE databases, and industry threat reports. • Update vendor risk assessments and enforce response timelines for critical third-party patching. • Coordinate with service providers to verify remediation actions or elevate high-risk exposures through contractual or governance channels.	[DISCOVER] Detect vulnerabilities and threats linked to third-party dependencies	Blockchain App (Custodial)
				• [CONTRIBUTE] Help shape industry standards for secure custodial operations and regulatory alignment	• Participate in industry working groups and standards bodies (e.g., FATF, ISO, CCSS, Global Digital Finance) to shape security, custody, and compliance frameworks. • Propose improvements to AML/CFT practices, wallet security protocols, and transaction monitoring standards based on operational experience. • Collaborate with regulators, audit firms, and technical experts to align evolving standards with both risk management and innovation. • Contribute to open-source security benchmarks and key management best practices, including cold storage, MPC, and HSM implementations. • Share lessons learned from audits, incident response, or threat intelligence to help raise the baseline across the industry.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [USE STANDARD] Implement and uphold recognized security and compliance standards	• Adopt custody and wallet security standards. • Implement regulatory compliance frameworks, including KYC, AML/CFT controls, transaction monitoring, and recordkeeping protocols. • Align infrastructure and operations with technical best practices, such as secure API design, key management (e.g., MPC/HSM), and cold/hot wallet segregation. • Integrate risk management procedures that comply with jurisdiction-specific licensing, reporting, and audit obligations. • Regularly update systems and procedures to stay aligned with evolving global standards and supervisory guidance. • Participate in certification or audit programs to verify ongoing compliance and demonstrate accountability to users and regulators.	[END]	-
Affiliated Third Parties	Logistics and Supply Chain Consumers / Relying Party		• Traditional Finance operators • Financial Institutions	• [ASSESS] Monitor dependency risks across blockchain and internet service providers	• Evaluate the security and reliability of third-party blockchain vendors, including L1/L2 protocols, node operators, and wallet providers. • Assess exposure to internet infrastructure risks, such as DNS or routing failures, that could disrupt user-facing services or on-chain transaction flow. • Review threat intelligence and incident reports involving vendors or infrastructure partners to update internal risk assessments.	[ANALYZE]	Consumers/Relying Party Financial Regulatory Authority

		• Marketplaces • Investors / Consumers		• Incorporate blockchain-specific vendor risks into enterprise supply chain and technology risk management frameworks. • Monitor service-level agreements and vendor performance for signs of degraded security posture, compliance failure, or cascading risks. • Adapt procurement and integration decisions based on known vulnerabilities or systemic issues in third-party blockchain platforms.	[END]	
					[CONTRIBUTE]	Consumers/Relying Party
			• [ANALYZE] Assess third-party contractual and operational risk	• Analyze shared security incidents or vulnerabilities involving third-party blockchain protocols/apps to assess contract compliance and service continuity risk. • Map disclosed threats to specific obligations (e.g., uptime SLAs, audit rights, security controls) to evaluate vendor reliability and liability. • Review whether third-party weaknesses could result in financial loss, regulatory exposure, or reputational damage. • Coordinate with procurement, risk, and legal teams to update internal risk ratings and determine if contractual amendments or contingencies are needed. • Assess concentration risk and systemic exposure when multiple dependencies are linked to the same vulnerable third-party services. • Use findings to inform future vendor selection, onboarding procedures, and third-party monitoring frameworks.	[SHARE]	Consumers/Relying Party
			• [SHARE] Communicate contract-related risk findings to third-party blockchain protocols/apps	• Feed analysis results into due diligence and onboarding processes for new blockchain or infrastructure partners. • Share risk findings related to vendor performance, including SLA breaches, recurring outages, or inadequate incident response. • Notify vendors of compliance gaps or security issues that may trigger contractual obligations (e.g., audit clauses, data breach reporting). • Escalate material risks that could impact contractual continuity (e.g., vulnerability disclosures, protocol instability, governance failures). • Use contractual channels (e.g., governance boards, formal communications, legal notices) to document risk positions and request action. • Coordinate disclosures with procurement and legal teams to inform renegotiation, due diligence updates, or exit strategies. • Promote transparency and accountability through structured, traceable communication that links risk to specific contract terms or expectations.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
			• [CONTRIBUTE] Provide user-driven input to the development of blockchain security standards	• Participate in industry forums or working groups to represent consumer and enterprise requirements. • Contribute insights based on operational pain points, vendor oversight challenges, and risk management expectations. • Advocate for security and reliability standards that support institutional-grade custody, AML/CFT alignment, and incident response clarity. • Support development of interoperability and disclosure standards to facilitate due diligence, integration, and monitoring across blockchain services. • Push for inclusion of contractual transparency and accountability mechanisms within emerging governance or trust frameworks. • Validate whether proposed standards are practical and adoptable for regulated institutions and large-scale consumer platforms.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [USE STANDARD] Adopt established standards in evaluating and engaging blockchain service providers	• Apply technical and security standards during vendor due diligence and procurement processes. Require third-party providers to comply with industry-accepted protocols for custody, key management, transaction integrity, and service uptime. • Align internal risk assessment and oversight practices with shared frameworks for vulnerability disclosure, incident reporting, and operational resilience. • Adopt regulatory standards (e.g., FATF, MiCA, or FSB guidelines) for KYC/AML, data handling, and financial integrity when engaging blockchain services. • Incorporate best practices into onboarding and SLA agreements to enforce baseline security and interoperability across blockchain-based offerings. • Update compliance procedures as standards evolve, ensuring long-term alignment with sector-wide expectations and regulatory guidance.	[END]	-

	Security Expert	• Security Practitioners • Ethical hackers • Individual security experts • Web3 security audit firms • Security Researchers	• [DISCOVER] Detect security flaws in blockchain protocols, apps, and networks	• Perform vulnerability assessments of smart contracts, bridges, and DeFi protocols using static and dynamic analysis techniques. • Conduct penetration tests and audits on blockchain applications, RPC endpoints, and validator infrastructure. • Use fuzzing, formal verification, or manual review to detect edge-case bugs, logic flaws, and unsafe interactions. • Analyze dependency chains and third-party integrations (e.g., oracles, wallet connectors, libraries) for systemic weaknesses. • Submit findings via responsible disclosure processes or participate in bug bounty programs to facilitate remediation. • Continuously monitor blockchain networks and mempools for signs of emerging exploit patterns or unpatched vulnerabilities.	[ALERT]	Security Expert
			• [DISCOVER] Detect emerging threats targeting blockchain systems and users	• Monitor blockchain networks, mempools, and bridges for signs of ongoing exploits, frontrunning, and manipulation. • Identify attacker behaviors such as phishing, drainer contracts, malicious token issuance, and wallet signature abuse. • Reverse-engineer malware or malicious contracts used in attacks to understand methods and build defenses. • Track threat actors and exploit campaigns through on-chain analysis, off-chain telemetry, and open-source intelligence. • Detect novel attack vectors in evolving protocol features, MEV mechanisms, or cross-chain interactions. • Collaborate with industry alliances and CERTs to validate, escalate, and share threat intelligence in a coordinated manner.	[REPORT]	Security Expert
			• [ALERT] Notify affected project maintainers of identified vulnerabilities or threats	• Initiate responsible disclosure by reporting vulnerabilities directly to project maintainers through designated security channels. • Alert protocol developers and service operators when real-time threats (e.g., ongoing exploits, drainer attacks) are detected in the wild. • Provide technical documentation and proof-of-concept evidence to help project maintainers understand, reproduce, and fix the issue efficiently. • Coordinate disclosure timelines and confidentiality, especially for critical or zero-day vulnerabilities, to allow safe remediation. • Engage with vendor security teams to track resolution status and offer support in patch validation or retesting.	[ESCALATE]	Security Expert
					[REMEDIATE]	Blockchain Protocol Blockchain Network Blockchain App
			• [ESCALATE] Disclose unresolved vulnerabilities or threats when vendors fail to remediate	• Track remediation status of disclosed issues and document vendor inaction, especially for high-severity risks. • Follow responsible disclosure timelines, and if unheeded, initiate public or community-led disclosure in accordance with ethical norms. • Report unremediated threats or vulnerabilities to third-party coordination entities (e.g., CERTs, Crypto-ISACs, bug bounty platforms). • Warn affected users or industry participants when continued vendor negligence presents imminent risk to funds, data, or protocol integrity. • Publish post-disclosure findings in a transparent and constructive manner, helping others recognize and respond to unresolved threats. • Support broader calls for accountability, including recommendations for vendor delisting, patch enforcement, or governance action.	[WARN]	Gov. Lab/Response Team Crypto-focused ISAC
			• [REPORT] Disclose identified vulnerabilities to the broader ecosystem through trusted channels	• Submit verified vulnerability data to threat intelligence platforms, bug bounty aggregators, or public security forums. • Publish vulnerability write-ups or technical advisories post-remediation, including reproduction steps and mitigation guidance. • Contribute anonymized vulnerability insights to sector-wide knowledge bases.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
					[ASSESS]	Blockchain Protocol

				• Engage in postmortems or collaborative reviews with other security researchers and developers to contextualize the vulnerability.		Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authority
			• [REPORT] Share actionable threat intelligence with the broader blockchain ecosystem	• Report indicators of compromise (IOCs) such as malicious contract addresses, phishing URLs, drainers, and exploit toolkits. • Share threat reports or technical analyses of attack campaigns observed in the wild, including TTPs (tactics, techniques, procedures). • Contribute intelligence to sector-wide platforms, including threat intel feeds, crypto-focused ISACs, and community alert channels. • Alert peers to new forms of social engineering or signature spoofing affecting users or validators. • Participate in real-time coordination during active incidents, contributing to rapid containment, attribution, and mitigation efforts.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
					[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [ASSESS] Evaluate security risks across blockchain systems using discovered or reported vulnerability or threat information	• Perform contextual risk assessments for protocols, dApps, bridges, or node clients based on newly disclosed or observed vulnerabilities. • Evaluate potential exploitability and impact severity across different threat scenarios, including privilege escalation, fund loss, or denial of service. • Correlate threat intelligence with architectural weaknesses in deployed systems (e.g., upgradeability flaws, access control gaps, oracle dependencies).	[DISCOVER]	Security Expert
					[ANALYZE]	Consumers/Relying Party

				• Publish risk analyses or severity ratings to guide urgency of remediation and stakeholder awareness. • Update risk models in response to ongoing research, incident reports, or collaborative investigations with other security teams.		Security Expert Auditor Blockchain Analytics Firm
						Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
					[END]	-
					[CONTRIBUTE]	Security Expert
			• [ANALYZE] Examine vulnerabilities and threats to produce actionable insights on cybersecurity for the blockchain ecosystem	• Perform in-depth analyses of disclosed or discovered issues, including vulnerability classification, exploit mechanics, and root cause identification. • Aggregate findings across audits, disclosures, and incidents to identify trends in attacker behavior, codebase weaknesses, or dependency risks. • Publish threat landscape reports, vulnerability taxonomies, or security deep dives to inform protocol teams, developers, and infrastructure providers. • Develop tools, checklists, or frameworks based on recurring issues to help others prevent similar vulnerabilities.	[SHARE]	Security Expert
			• [SHARE] Disseminate analytical findings and on vulnerabilities and threats to inform and strengthen the blockchain ecosystem	• Publish technical analyses, postmortems, and attack deconstructions to explain the cause, method, and impact of discovered security issues. • Share synthesized threat patterns and vulnerability trends through blogs, reports, webinars, or academic papers to raise ecosystem-wide awareness. • Contribute to public repositories or threat intelligence platforms that aggregate lessons learned from security incidents. • Collaborate with developer communities, ISACs, and audit networks to ensure insights are actionable and widely disseminated. • Translate findings into recommendations that support safer contract development, protocol governance, and vendor selection. • Facilitate learning loops across incidents, ensuring similar vulnerabilities are not repeated across projects or ecosystems.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit

			• [CONTRIBUTE] Provide expert input to security standard development for blockchain systems	• Participate in standards bodies and working groups to shape specifications on secure development, disclosure, key management, and network security. • Draft or review proposed standards and guidelines, offering feedback based on empirical vulnerabilities and threat research. • Recommend technical controls and protocols informed by real-world exploits and audit findings across blockchain systems. • Promote alignment between standards and open-source security practices, ensuring that frameworks reflect actual developer needs and tooling.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Build tools and frameworks that embed and operationalize security standards	• Develop auditing tools, static analyzers, and testing suites that enforce smart contract or protocol-level security standards. • Create developer frameworks and security libraries that abstract complex controls in line with recognized standards. • Embed standards into continuous integration/continuous deployment pipelines, enabling automated compliance checks. • Maintain open-source templates and checklists that reflect evolving guidelines for smart contract deployment, node configuration, or wallet architecture. • Collaborate with protocol teams and foundations to integrate security standards into ecosystem tooling (e.g., SDKs, deployment dashboards). • Educate developers and auditors through documentation, training material, or plug-and-play infrastructure aligned with industry benchmarks.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party
	Auditor	• Auditors • Audit firms	• [DISCOVER] Detect security weaknesses in client systems through formal audits	• Conduct structured security audits of smart contracts, dApps, custody systems, and validator/node infrastructure. • Perform code reviews, threat modeling, and static/dynamic analysis to uncover logic errors, access control issues, and unsafe patterns. • Simulate real-world attack scenarios to assess exploitability and severity of potential vulnerabilities. • Continuously update auditing methodologies to reflect evolving threats, standards, and tooling in the blockchain ecosystem.	[ALERT]	Auditor
			• [DISCOVER] Detect security threats relevant to client systems and operations	• Identify relevant threat actors, tactics, and techniques that could target the client's protocol, dApp, or infrastructure. • Assess exposure to known attack vectors, such as MEV exploitation, phishing, front-running, oracle manipulation, or governance attacks. • Analyze client architecture for real-world threat impact, including how external events or dependencies may introduce risk. • Perform red-team simulations or adversarial testing to validate how threats could bypass current defenses.	[REPORT]	Auditor
			• [ALERT] Notify clients of identified security vulnerabilities and potential threats	• Deliver comprehensive audit reports that outline discovered vulnerabilities and threats. • Schedule debrief sessions or real-time escalation calls to ensure critical findings are communicated before deployment or during time-sensitive operations. • Provide remediation guidance tailored to the client's infrastructure, development resources, and project timeline. • Maintain secure communication protocols and confidentiality throughout the reporting process to protect client interests while encouraging prompt action.	[ESCALATE]	Auditor
					[REMEDIATE]	Blockchain Protocol Blockchain Network Blockchain App
			• [ESCALATE]	• Track the remediation status of high-risk issues and maintain internal logs of client responses and timelines.	[WARN]	Gov. Lab/Response Team Crypto-focused ISAC

			Initiate responsible disclosure when critical vulnerabilities or threats that could impact the broader ecosystem are not remediated	• Initiate escalation workflows if the client does not address critical issues, including notifications to project stakeholders, bug bounty platforms, or third-party disclosure coordination groups (e.g., CERTs, Crypto-ISAC). • Coordinate with legal or ethical advisory teams to ensure any escalation complies with contractual obligations and professional standards. • Provide anonymized or redacted summaries when appropriate, to warn the broader community of unresolved systemic risks without disclosing sensitive project details.		
			• [REPORT] Alert clients and relevant stakeholders about potential threats identified	• Notify clients of suspicious contract behavior, deployment anomalies, or ecosystem-wide attack indicators uncovered during security assessments. • Identify and flag potential protocol-level risks that could cascade beyond the audited system. • Recommend immediate defensive actions such as pausing contracts, warning users, or isolating risky integrations based on observed anomalies or disclosed exploits. • Share anonymized intelligence with threat-sharing alliances, open-source security communities, or CERTs, when threats pose systemic or cross-protocol risk. • Collaborate with affected parties or ecosystem partners to validate findings and support collective remediation.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
					[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
			• [ASSESS] Conduct security audits and provide advisory services to clients	• Perform security audits of blockchain protocols, smart contracts, custodial systems, wallets, bridges, and node infrastructure. • Evaluate cryptographic implementations, transaction validation logic, access controls, and operational security measures. • Conduct both manual code review and automated scanning using proprietary or open-source tools. • Test for known vulnerability classes (e.g., reentrancy, underflow, improper input validation) and assess business logic flaws. • Provide audit reports with risk severity ratings, proof-of-concept exploits, and prioritized mitigation steps. • Follow up with re-audits to confirm remediation of vulnerabilities and compliance with findings.	[DISCOVER]	Auditor
					[ANALYZE]	Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm
						Academia Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit Gov. Lab/Response Team Crypto-focused ISAC
					[END]	-
					[CONTRIBUTE]	Auditor

			• [ANALYZE] Examine trends in vulnerabilities and threats to improve audit practices	• Aggregate findings across engagements to identify systemic flaws in commonly used libraries, contract patterns, or protocol architectures. • Analyze on-chain exploit incidents and correlate them with audit gaps or missed risk vectors to improve test coverage. • Review threat intelligence sources, postmortems, and disclosed attack vectors to align audit scopes with real-world risks.	[SHARE]	Auditor
			• [SHARE] Publish insights and updates to audit methodologies	• Publish technical articles, audit retrospectives, and lessons-learned reports summarizing key vulnerabilities and how they were detected. • Update internal audit processes and checklists based on the above analyses to ensure future assessments are more robust and forward-looking. • Release updates to open-source auditing tools, severity classification systems, or procedural templates, encouraging consistency across firms. • Present findings at security conferences, working groups, or standards meetings to facilitate knowledge exchange and community awareness. • Create educational content (e.g., guides, documentation, interactive labs) that reflects updated audit frameworks and supports secure development.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
			• [CONTRIBUTE] Offer expert input into the development of security and auditing standards	• Join working groups or industry alliances to co-develop or review proposed standards. • Provide feedback on draft specifications related to contract security, key management, validator operations, or incident disclosure procedures. • Translate field experience into normative language that ensures standards address practical issues. • Advocate for standards that support auditability, transparency, and user protection across all blockchain layers.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Align audit frameworks with evolving standards	• Integrate technical benchmarks into audit templates, checklists, and reporting formats. • Regularly review new or revised standards and update internal frameworks to reflect changes in terminology, scoring, or scope. • Educate clients on the relevance of adopted standards and encourage their implementation in both design and deployment stages. • Collaborate with standard-setting bodies to validate whether their frameworks are being correctly and consistently applied across audit firms.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party
	Insurance Company	• Insurance Company (Cyber Insurance)	• [ASSESS] Evaluate cybersecurity risks and underwrite blockchain-related insurance coverage	• Conduct security risk assessments based on technical audits, operational maturity, incident response readiness, and third-party exposure. • Evaluate client adherence to best practices in areas such as smart contract deployment, key management, off-chain infrastructure security, and disclosure procedures. • Define coverage parameters based on risk factors including protocol complexity, DeFi composability risks, attack history, and on-chain liquidity exposure. • Provide tiered policies that reflect risk segmentation and reward adherence to security standards and audit results. • Collaborate with brokers and clients to tailor policies for specific blockchain use cases (e.g., validator slashing, bridge failure, or oracle manipulation). • Use breach data and threat intelligence to continuously refine underwriting models and payout eligibility criteria.	[END] [CONTRIBUTE]	- Insurance Company
			• [CONTRIBUTE] Help shape cybersecurity and risk standards through insurance industry expertise	• Participate in standards-setting organizations or working groups to offer insurance-specific input on security and disclosure practices. • Advise on risk quantification models, helping define how vulnerabilities, threat exposure, and incident response maturity should factor into broader resilience standards. • Contribute incident trend data to shape realistic benchmarks and coverage recommendations across blockchain services.	[STANDARDIZE]	International Standards Organization National Standards Organization

				• Validate proposed standards based on how effectively they reduce claim frequency or severity in real-world insurance portfolios. • Bridge gaps between technical standards and regulatory insurance frameworks, supporting alignment across risk, compliance, and operational domains.		
			• [PROMOTE STANDARD] Reflect standards on cyber insurance coverage and pricing	• Offer premium discounts or enhanced coverage to clients that adopt verified standards or undergo independent audits. • Incorporate security framework compliance into risk scoring algorithms and underwriting checklists. • Reward proactive threat detection, secure coding practices, and standardized disclosure processes through more favorable policy conditions. • Adjust pricing models dynamically based on adherence to updated best practices, threat intelligence signals, and validated security implementations. • Use standards as a benchmark to determine insurability of new blockchain services or protocols with limited operational history.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party
	Blockchain Analytics Firm	• Blockchain Analytics firms (Chainalysis, Elliptic, etc)	• [DISCOVER] Monitor blockchain networks to detect illicit activity and threat patterns	• Continuously monitor public blockchain data using clustering algorithms, labeling heuristics, and address attribution techniques to detect suspicious activity. • Identify behavioral patterns associated with mixers, ransomware payments, phishing campaigns, and exploit laundering. • Flag addresses or contract interactions linked to known threat actors, sanctioned entities, or compromised protocols. • Integrate off-chain data (e.g., metadata, social signals, open-source intelligence) to contextualize on-chain behavior and enhance detection accuracy. • Provide clients with risk-scored address profiles, wallet attribution reports, and activity heatmaps based on real-time observations.	[REPORT]	Blockchain Analytics Firm
			• [REPORT] Notify clients and the ecosystem of potential threats based on detected anomalies or illicit behavior	• Send real-time alerts when monitored wallets interact with high-risk addresses or when new laundering patterns emerge. • Notify clients (blockchain apps/protocols) and the ecosystem that stolen funds from exploits are routed through their platforms. • Provide actionable intelligence for freezing, monitoring, or rejecting suspicious transactions. • Alert compliance teams to activity involving blacklisted or watchlisted entities. • Offer contextual risk explanations to help clients understand the urgency and type of threat.	[DISCLOSE]	Gov. Lab/Response Team Crypto-focused ISAC
					[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
			• [ASSESS] Provide advisory services and forensic support to clients	• Conduct wallet tracing investigations and produce forensic reports for litigation, incident response, or law enforcement support. • Help institutions assess counterparty risk and exposure to high-risk DeFi contracts or wallets. • Provide compliance assessments and AML/CFT alignment recommendations based on transaction history and behavior analysis. • Assist in internal investigations for insider trading, employee misuse, or operational security breaches. • Support pre-listing risk evaluations of tokens or projects for centralized exchanges and custody platforms.	[DISCOVER]	Blockchain Analytics Firm
					[ANALYZE]	Blockchain Analytics Firm
					[END]	

					[CONTRIBUTE]	Blockchain Analytics Firm
			• [ANALYZE] Continuously examine threat trends to improve monitoring systems and inform clients	• Track how laundering behavior evolves across bridges, DEXs, mixers, and cross-chain protocols. • Analyze smart contract exploits to identify typical laundering routes and asset obfuscation methods. • Develop behavioral models to identify new scam types, exploit trends, or manipulation tactics. • Feed analytics results into dashboard updates, new clustering models, and false-positive reduction algorithms. • Brief clients on ecosystem-wide threat evolution via newsletters, webinars, and private consultations.	[SHARE]	Blockchain Analytics Firm
			• [SHARE] Disseminate aggregated threat intelligence and forensic insights to the ecosystem	• Publish quarterly or ad hoc threat intelligence reports outlining trends in illicit finance, attack patterns, and laundering methods. • Share wallet blacklists, scam contract addresses, and typologies with DeFi protocols, regulators, and law enforcement. • Contribute to multi-stakeholder platforms like TRISA, Crypto-ISAC, or BGIN with anonymized case studies. • Offer workshops or public briefings to regulators, exchanges, and compliance professionals based on forensic learnings. • Maintain open APIs or feeds for real-time threat data integration by ecosystem participants.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Auditor Insurance Company Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [CONTRIBUTE] Participate in development of risk classification and compliance standards	• Provide expert input into regulatory consultations (e.g., FATF Travel Rule) and blockchain-specific frameworks. • Help define categories and thresholds for wallet risk scoring, typology classification, and illicit activity detection. • Recommend best practices for suspicious activity detection, reporting thresholds, and anomaly escalation protocols. • Support interoperability standards for cross-border data exchange among VASPs and compliance systems. • Contribute to the development of incident disclosure and analytics integrity standards across firms.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Embed accepted standards into monitoring tools and client-facing products	• Implement FATF-aligned risk scoring frameworks and Travel Rule compliance modules in analytics dashboards. • Embed industry-approved watchlists, threat classification models, and KYC-linked wallet identifiers into client-facing systems. • Customize alerts and threshold logic to match evolving regulatory guidance and incident response expectations. • Provide automated reporting exports that align with standardized reporting formats. • Ensure traceability, auditability, and explainability of risk ratings and analysis output in line with transparency standards.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App

						Consumers/Relying Party
Independent / Public Third Parties	Certification Body	• Security Certification schemes • Testing/Certifying Lab (CC, CMVP)	• [ASSESS] Evaluate security posture and issue certifications based on defined standards	• Perform structured evaluations of blockchain protocols, wallets, custody systems, and smart contract platforms against established technical and procedural criteria. • Use repeatable testing methodologies to assess system design, implementation security, key management, access controls, and vulnerability response processes. • Analyze system-specific risks as part of the certification process, including dependency risk, third-party integrations, and operational maturity. • Issue certifications upon successful compliance, or provide corrective action recommendations when gaps are identified. • Maintain audit trails, version control, and re-evaluation schedules to ensure certified entities remain compliant over time. • Support public transparency by listing certified products or providers and the scope of certification awarded.	[END]	
					[CONTRIBUTE]	Certification Body
			• [CONTRIBUTE] Contribute to standards	• Participate in standards-setting initiatives to define or refine frameworks related to blockchain security, custody assurance, or smart contract integrity. • Offer feedback based on certification data—highlighting areas where standards are ambiguous, insufficiently tested, or misaligned with field conditions. • Propose new criteria for emerging blockchain components that lack formal guidance. • Help align terminology and assessment criteria across different standards bodies and certification schemes to reduce duplication and enhance clarity. • Contribute anonymized results or systemic findings to inform cross-industry discussions on assurance models and threat resilience.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Integrate standards into certification criteria	• Map industry-recognized standards to specific controls, evaluation procedures, and test cases used in certification. • Update certification checklists and scoring rubrics to reflect changes in standards, threat models, or consensus-driven best practices. • Define pass/fail thresholds and evaluation protocols that ensure consistent interpretation of abstract standards across auditors and labs. • Train assessors and auditors to apply updated certification criteria, ensuring repeatability and fairness in evaluations. • Encourage adoption by publishing updated frameworks, crosswalks between standards, and user guides for applicants preparing for certification.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party
	Academia	• Academia (Academic Peer-Reviewed Papers) • Contribute to standard developmentAcademia (Sponsored Research)	• [ANALYZE] Conduct academic research on security vulnerabilities and emerging threats	• Design and conduct empirical studies on attack surfaces, vulnerability propagation, smart contract flaws, validator behavior, and Layer 2 security properties. • Develop formal models, simulations, or datasets to evaluate protocol resilience, game-theoretic dynamics, and adversarial behavior in decentralized systems. • Investigate real-world incidents or protocol failures to draw generalizable lessons and identify systemic risks. • Publish peer-reviewed research papers, whitepapers, and working papers that contribute to the body of knowledge on blockchain security. • Collaborate with industry partners through sponsored research or joint projects to study cutting-edge issues • Provide an academic evidence base that informs policy, tooling, and engineering decisions in the blockchain ecosystem.	[SHARE]	Academia
			• [SHARE] Disseminate research findings and build capacity through education and public engagement	• Deliver lectures, seminars, and continuing education programs focused on blockchain security, cryptography, and protocol analysis. • Publish research in accessible formats (e.g., blog posts, public datasets, explainer videos) to broaden reach beyond academia. • Organize or contribute to open-source curriculum and online courses on secure smart contract development, vulnerability discovery, and threat modeling. • Mentor students and practitioners through research labs, hackathons, or summer schools that focus on practical security experimentation. • Participate in multi-stakeholder workshops and conferences to bridge academic theory with real-world engineering practice.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App

				• Encourage open dialogue by sharing early-stage research and soliciting community feedback for refinement and co-creation.		Consumers/Relying Party Security Expert Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [CONTRIBUTE] Shape the development of technical and policy standards through research-based insights	• Participate in technical committees, standardization bodies, and open working groups to draft and review blockchain-related security standards. • Contribute empirical data, attack taxonomies, and modeling frameworks to support well-justified benchmarks and control definitions. • Review or co-author sections of security, interoperability, or privacy standards based on experimental or theoretical findings. • Advocate for standards that reflect cryptographic soundness, protocol diversity, and adversarial awareness. • Facilitate alignment between academic frameworks (e.g., formal verification, composability models) and practical standards adopted by developers and auditors. • Act as a neutral voice mediating between industry, regulators, and researchers to ensure that standards reflect both feasibility and long-term resilience.	[STANDARDIZE]	International Standards Organization National Standards Organization
	Law Enforcement Authorities	• Law Enforcement Authorities (Europol, INTERPOL, FBI, etc)	• [ASSESS] Conduct investigations and take enforcement action based on gathered evidence	• Investigate crimes such as fraud, hacking, money laundering, terrorism financing, or sanctions evasion involving digital assets. • Use blockchain analytics tools, subpoenas, surveillance, and international coordination to trace fund flows and identify suspects. • Arrest individuals or dismantle networks involved in criminal schemes once probable cause and jurisdictional requirements are met. • Coordinate with prosecutors and judicial bodies to ensure evidence integrity and due process. • Protect victims and recover assets through seizure and restitution efforts where legally possible.	[ANALYZE]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Blockchain Analytics Firm
						Financial Intelligence Unit Crypto-focused ISAC
					[END]	
					[CONTRIBUTE]	Law Enforcement Authority
			• [ANALYZE] Interpret threat intelligence to determine enforcement strategy and legal pathways	• Review threat reports from analytics firms, exchanges, and other agencies to validate claims and assess credibility. • Cross-reference wallet activity, KYC data, or on-chain behavior with case databases to confirm illegal conduct. • Determine whether domestic criminal statutes (e.g., wire fraud, money laundering) or international legal frameworks (e.g., MLATs, INTERPOL notices) apply. • Assign investigative leads to relevant units (e.g., cybercrime, financial crimes, terrorism financing) based on case characteristics.	[SHARE]	Law Enforcement Authority

				• Consider diplomatic, reputational, and privacy implications in cross-border enforcement decisions.		
			• [SHARE] Disseminate enforcement strategies and lessons learned to support ecosystem security	• Financial	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party Security Expert Blockchain Analytics Firm Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [CONTRIBUTE] Help shape international and national AML/CFT standards through investigative insight	• Collaborate with standard-setting bodies like FATF, Egmont Group, or Interpol's Financial Crime Division to provide feedback on draft AML/CFT guidance. • Recommend threshold updates, typologies, and red flags based on case investigations involving mixers, privacy coins, NFTs, or DeFi protocols. • Contribute anonymized data on trends in laundering behavior, transaction structuring, and fund obfuscation techniques. • Support efforts to define traceability requirements, jurisdictional cooperation mechanisms, and enforcement data-sharing models. • Align law enforcement priorities with evolving compliance frameworks to improve the effectiveness of reporting and supervision tools used by VASPs and financial institutions.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Integrate standard into enforcement	• Incorporate international AML/CFT frameworks into enforcement workflows and investigations involving virtual assets. • Align investigative documentation and evidence collection with standardized typologies, ensuring legal interoperability and case transferability across borders. • Apply technical standards (e.g., blockchain auditability, smart contract forensics, metadata tagging) when analyzing on-chain evidence or preparing judicial reports. • Use standards-based checklists and thresholds (e.g., transaction patterns, wallet behaviors) to guide the classification of suspicious activity and trigger escalation procedures. • Collaborate with compliance teams, analytics providers, and regulators to ensure enforcement activity supports industry adoption of shared security and reporting standards. • Train officers and investigators on how to interpret and operationalize technical and regulatory standards within blockchain-specific cases.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party
	Financial Regulatory Authorities	• Financial Regulatory Authorities (SEC, CFTC, JFSA, MAS, FCA, BaFin, AMF, etc) Courthouse and attorney (for trial)	• [ASSESS] Evaluate risks of blockchain-related projects from the lens of customer protection, crime prevention, and financial stability	• Monitor blockchain projects platforms for vulnerabilities, exploit attempts, or market distortions that may result in investor harm, systemic disruption, or legal violations. • Assess the impact of technical or governance failures—such as protocol bugs, asset freezes, or stablecoin depegging—on consumer protection and market confidence. • Evaluate business models, control structures, and resilience of blockchain entities to determine potential exposure to financial crime, cyber risk, or operational fragility. • Respond to reports from external stakeholders by conducting supervisory reviews or initiating investigatory processes. • Engage internal or court-led risk assessments during regulatory action, enforcement proceedings, or litigation involving blockchain service providers.	[ANALYZE]	Blockchain Protocol Blockchain App (Custodial, VASPs) Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm

						Financial Regulatory Authority Crypto-focused ISAC
					[END]	
					[CONTRIBUTE]	Financial Regulatory Authority
			• [ANALYZE] Conduct cross-sectoral risk analysis of blockchain market trends and systemic vulnerabilities	• Aggregate incident and exposure data across exchanges, stablecoins, lending protocols, and custody providers to evaluate market-wide trends. • Analyze network structures, liquidity dependencies, or protocol interconnections to identify systemic points of failure or contagion risk. • Assess the resilience of blockchain market infrastructure under stress scenarios (e.g., depegging, bridge failure, insolvency events). • Map illicit activity pathways and regulatory arbitrage patterns to strengthen supervision, cross-border enforcement, and policy design. • Produce internal or published reports that synthesize technical findings into actionable financial policy perspectives.	[SHARE]	Financial Regulatory Authority
			• [SHARE] Share recommendations based on analysis	• Release public statements, circulars, or advisories on risks associated with algorithmic stablecoins, tokenization, staking services, or custody controls. • Engage in bilateral or multilateral dialogue with foreign regulators and institutions to share policy responses and enforcement trends. • Publish lessons learned and thematic reviews following high-profile incidents, enforcement actions, or supervisory assessments. • Provide technical recommendations to industry bodies for implementing robust internal controls and user protection mechanisms. • Conduct outreach sessions with firms and developers to explain compliance expectations and emerging regulatory positions.	[ASSESS]	Blockchain Protocol Blockchain App (Custodial, VASPs)
						Consumers/Relying Party Security Expert Auditor Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [CONTRIBUTE] Help shape international and national financial regulatory standards based on analysis	• Participate in standard-setting bodies such as the FSB, IOSCO, FATF, and BIS to shape digital asset policy frameworks. • Propose revisions to existing rules based on gaps identified in cross-border DeFi activity, exchange supervision, or token classification. • Incorporate courtroom precedents and enforcement decisions into the evolving legal interpretation of digital assets. • Recommend regulatory definitions and disclosure requirements that balance innovation, investor protection, and financial stability. • Support coordination across law enforcement, financial supervisors, and judiciary to harmonize oversight and prosecution of blockchain-related misconduct.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Integrate standard into regulations	• Codify international standards (e.g., FATF, IOSCO, ISO) into local regulations, licensing regimes, and operational guidelines. • Require financial institutions and blockchain service providers to adopt certified security or audit standards as a condition of authorization. • Align enforcement actions and legal rulings with recognized benchmarks to ensure consistency and signal legal clarity.	[USE STANDARD]	Blockchain Protocol Blockchain App (Custodial, VASPs)

				• Incorporate technical standards into prudential supervision tools, data collection templates, and stress testing methodologies. • Encourage industry-wide adoption of compliance frameworks by rewarding adherence in sandbox approvals, fintech licensing, or regulatory relief schemes.		Consumers/Relying Party
	Financial Intelligence Units	• Financial Intelligence Units (FinCEN, FIU-IND, NCA, Egmont Group, etc/FATF)	• [ASSESS] Investigate blockchain-related threats to financial integrity and national security	• Investigate suspicious blockchain transactions flagged through STRs/SARs or reported by analytics firms, VASPs, or law enforcement. • Trace on-chain fund flows involving mixers, unhosted wallets, cross-chain bridges, or privacy-enhancing technologies. • Assess links between blockchain activity and known threat actors, including sanctioned individuals, fraud rings, and terror financing networks. • Validate and escalate high-risk cases to law enforcement, customs, or tax authorities, based on financial intelligence and jurisdictional relevance. • Coordinate domestic and cross-border investigative efforts using FIU-to-FIU information exchange mechanisms and Egmont protocols.	[ANALYZE]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Blockchain Analytics Firm
						Financial Intelligence Unit Crypto-focused ISAC
					[END]	
					[CONTRIBUTE]	Financial Intelligence Unit
			• [ANALYZE] Monitor and interpret threat trends across jurisdictions and technologies	• Aggregate and review suspicious activity reports (SARs) involving virtual assets to identify high-risk patterns or technologies. • Analyze changes in laundering techniques, such as layering via DEXs, misuse of NFTs, or repeated bridge hopping. • Correlate blockchain intelligence with traditional financial data, identifying hybrid laundering networks or jurisdictional arbitrage. • Track geographic trends in illicit activity and service provider risk, supporting risk-based supervision. • Publish internal risk assessments and typology reports to inform national policy, international partners, and law enforcement strategy.	[SHARE]	Financial Intelligence Unit
			• [SHARE] Disseminate threat trend insights with domestic and global partners	• Issue typology reports, advisories, and risk notices to blockchain stakeholders on current laundering and threat patterns. • Contribute to Egmont Group or FATF reports summarizing crypto-specific financial crime intelligence. • Host public-private intelligence exchange forums that share strategic trends with regulated institutions and investigators. • Facilitate bilateral or multilateral briefings to ensure law enforcement and supervisory agencies stay aligned with threat evolution. • Collaborate with technology vendors to validate monitoring tools using real case studies and data-driven red flags.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert Blockchain Analytics Firm
						Certification Body Law Enforcement Authority

							Financial Regulatory Authority Financial Intelligence Unit
				• [CONTRIBUTE] Inform international AML/CFT and virtual asset policy through threat-driven insights	• Provide anonymized case studies and typologies to standard-setting bodies shaping AML/CFT rules for blockchain services. • Propose revisions to Travel Rule, KYC requirements, or STR thresholds based on jurisdictional experience with evolving threats. • Advise on enforcement constraints, data-sharing challenges, and investigative bottlenecks related to digital assets. • Participate in FATF mutual evaluation processes and guidance development to ensure alignment between policy and financial intelligence. • Support the harmonization of definitions and risk categories across jurisdictions through Egmont coordination and FATF engagement.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [PROMOTE STANDARD] Recommend adoption of AML/CFT standards across blockchain stakeholders	• Advocate for integration of FATF-aligned controls in onboarding, monitoring, and reporting practices of crypto-related businesses. Recommend technical and procedural standards (e.g., address screening, source-of-funds verification, transaction monitoring) to reduce exposure to illicit activity. • Work with regulators and industry associations to promote standards implementation and threat awareness. • Encourage cross-sector capacity building and technical support to help smaller or emerging entities comply with AML/CFT expectations. • Support integration of standards into compliance tools and risk-scoring engines used by VASPs and compliance teams.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party
	Govt. Labs / Response Teams	• Vulnerability Coordination & Disclosure	• MITRE • national/institutional CERTs • National Labs	• [WARN] Notify blockchain project teams of discovered vulnerabilities or threats and recommend remediation	• Contact maintainers of blockchain protocols, dApps, wallets, or infrastructure when critical vulnerabilities or threat indicators are identified. • Provide detailed technical documentation and reproduction steps to help project teams validate and prioritize remediation. • Recommend timelines for disclosure and coordinate expectations under responsible disclosure frameworks. • Follow up on remediation progress and escalate when no action is taken on critical issues. • Where necessary, mediate disclosure across jurisdictions or when maintainers are pseudonymous or difficult to reach.	[REMEDIATE]	Blockchain Protocol Blockchain Network Blockchain App
						[DISCLOSE]	Govt. Labs / Response Teams Crypto-focused ISAC
				• [DISCLOSE] Publicly disclose confirmed vulnerabilities when remediation is complete or response is lacking	• Publish vulnerability advisories with technical detail, exploit context, and risk classification after a disclosure window closes. • Ensure disclosures align with CVE, CWE, or other industry-recognized identifiers for consistency. • Collaborate with open vulnerability databases and public registries to support global awareness. • Balance transparency with user protection by redacting sensitive details if exploitation risk remains high.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert
							Certification Body Law Enforcement Authority Financial Regulatory Authority
		• Cyber Threat Modeling		• [DISCLOSE]	• Publish technical reports and indicators of compromise (IOCs) related to blockchain-specific attack campaigns, malware, or exploit tooling.	[ASSESS]	Blockchain Protocol

		& Intelligence		Share confirmed threat intelligence with the public to raise ecosystem-wide awareness	• Share threat actor behavior patterns, such as laundering techniques, phishing methods, or smart contract manipulation strategies. • Issue alerts and advisories to VASPs, protocol maintainers, and public-private coalitions based on observed threats. • Coordinate cross-sector response during live incidents through joint bulletins and situational awareness briefings.		Blockchain Network Blockchain App Consumers/Relying Party Security Expert Blockchain Analytics Firm Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
		• Research & Tech Capability Development		• [ANALYZE] Conduct advanced testing, red teaming, and simulations to assess blockchain security	• Document proven defense-in-depth strategies for securing blockchain apps, protocols, and network. • Publish open-source detection logic, incident response workflows, and secure development guides. • Update public references (e.g., ATT&CK Navigator layers, CTI handbooks) for industry use. • Translate operational experience into actionable templates for sectoral use (e.g., wallet providers, bridges, L2s). • Apply frameworks like MITRE ATT&CK to classify observed attacker behavior. • Connect real-world incidents to known techniques and kill-chain stages. • Update threat models based on field data, simulations, and AARs (after-action reports).	[SHARE]	Govt. Labs / Response Teams
				• [SHARE] Share conducted research and developed tech capability	• Publish open-access whitepapers, frameworks, and tools that address threat modeling, attack detection, or remediation strategies. • Release open-source code for static analyzers, fuzzers, or on-chain monitoring tools developed during public research projects. • Present findings at technical conferences, standards workshops, and cybersecurity coalitions to ensure practical adoption. • Collaborate with academia, standards bodies, and private firms to refine methodologies through peer review.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party Security Expert Blockchain Analytics Firm Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
		• Public Infrastructure Support & Standards		• [CONTRIBUTE] Maintain a public repository of known vulnerabilities	• Operate or contribute to vulnerability databases (e.g., MITRE's CVE/NVD) specific to smart contracts, consensus mechanisms, or custody solutions. • Organize vulnerabilities using standard taxonomies (e.g., CWE, CAPEC) to promote consistent classification and awareness.	[STANDARDIZE]	International Standards Organization National Standards Organization

					• Document incident root causes and remediation outcomes for future reference by auditors, developers, and regulators. • Maintain living repositories with search and filtering tools to support proactive defense and education.		
				• [CONTRIBUTE] Provide expert input into the development of technical and cybersecurity standards	• Participate in national and international standardization bodies (e.g., NIST, ISO, IETF) shaping blockchain security norms. • Provide red-team-informed feedback on draft standards for key management, smart contract safety, or consensus validation. • Contribute formal models, risk scoring frameworks, and test datasets to support standard setting. • Act as conveners of expert working groups and cross-sector initiatives addressing open security gaps in blockchain systems.	[STANDARDIZE]	International Standards Organization National Standards Organization
				• [PROMOTE STANDARD] Build and deploy security tools that reflect emerging standards and best practices	• Develop and publish reference implementations of secure development workflows for smart contracts, custody systems, or validators. • Create automated tools that enforce compliance with emerging standards (e.g., audit checkers, key rotation verifiers). • Partner with ecosystem stakeholders to integrate these tools into CI/CD pipelines or developer toolkits. • Promote adoption through government-backed programs, public hackathons, and ecosystem support grants.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party
	Crypto-focused ISAC		• ISACs (CryptoISAC, JP-CryptoISAC, SEAL, etc)	• [DISCLOSE] Disseminate validated vulnerability and threat intelligence across the crypto sector	• Disseminate vetted vulnerability and threat reports submitted by members, security researchers, and partner organizations. • Standardize and distribute technical indicators (e.g., malicious addresses, phishing domains, contract exploits) through secure channels. • Coordinate timing of disclosures to allow impacted parties to remediate before public release where appropriate. • Help anonymize source attribution while maintaining context necessary for action. • Provide summaries and technical appendices for easy integration into security workflows and monitoring tools.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
							Consumers/Relying Party Security Expert Blockchain Analytics Firm
							Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
				• [ANALYZE] Evaluate remediation strategies and recommend practical responses for the industry	• Review how leading teams remediated vulnerabilities or threat incidents and identify replicable best practices. • Analyze the effectiveness of various mitigation techniques (e.g., circuit breakers, multisig upgrades, frontend shutdowns) across protocol types. • Develop playbooks or response templates tailored to common blockchain threat scenarios (e.g., governance attacks, bridge exploits). • Offer comparative assessments of post-incident outcomes to improve community response planning. • Help translate complex technical fixes into actionable guidance for non-technical project maintainers.	[SHARE]	Crypto-focused ISAC
				• [SHARE]	• Publish threat trend briefings, sector-level risk updates, or quarterly security landscape reports. • Facilitate closed-door workshops and roundtables to discuss emerging risks and successful mitigation efforts.	[ASSESS]	Blockchain Protocol Blockchain Network

			Circulate synthesized insights and post-incident learnings among ecosystem participants	• Share curated summaries of high-impact vulnerabilities, with remediation timelines and outcome analysis. • Create anonymized incident walkthroughs to foster learning without blame attribution. • Enable faster adoption of security insights across fragmented blockchain ecosystems. • Share vulnerability and threat intelligence shared by ISACs for different industries.		Blockchain App Consumers/Relying Party Security Expert Blockchain Analytics Firm Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [CONTRIBUTE] Maintain curated repositories of blockchain-specific threat intelligence and security practices	• Host searchable databases of historical incidents, known exploit patterns, and attacker TTPs (tactics, techniques, procedures). • Aggregate and organize member-contributed indicators, remediation tactics, and policy templates. • Tag entries by asset class (e.g., bridges, DEXs, NFT platforms) or technology stack to improve navigability. • Support structured data formats (e.g., STIX/TAXII) for machine-readable sharing across ISAC participants. • Ensure repositories are accessible to vetted members, law enforcement, and security researchers under a trust framework.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [CONTRIBUTE] Support the development of blockchain-specific security and risk standards	• Facilitate working groups to define or refine standards related to smart contract auditing, validator operations, and incident response. • Provide recommendations to standards bodies on sector-specific controls and disclosure frameworks. • Coordinate community feedback on draft proposals and surface field-tested insights. • Advocate for interoperability and threat reporting alignment across blockchains and service providers. • Support development of minimum baselines and certification criteria for secure blockchain system operation.	[STANDARDIZE]	International Standards Organization National Standards Organization
			• [PROMOTE STANDARD] Encourage the adoption of recognized standards across the crypto industry	• Offer onboarding resources, workshops, and assessments to help participants implement standards. • Develop tailored guidance showing how specific standards apply to smart contract teams, node operators, or custodial providers. • Host compliance readiness programs that connect members with auditors, tooling providers, and training resources. • Track and share adoption trends to encourage transparency and standard maturity across the ecosystem. • Incentivize conformance by linking standard usage to ISAC membership benefits, reputation mechanisms, or partner integrations.	[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party
	ISACs for different industries	• ISACs for other industries (FS-ISAC, NH-ISAC, etc)	• [ANALYZE] Analyze cross-industry vulnerability & threat intelligence	• Translate industry-specific cyber events into formats and language understandable across sectors. • Distribute alerts and situational awareness bulletins to other ISACs, government partners (e.g., CISA, ENISA), and critical infrastructure groups. • Host multi-sector calls and briefings during large-scale or systemic incidents (e.g., ransomware affecting multiple verticals). • Coordinate with sector risk management agencies (SRMAs) and national cyber centers to align messaging across industries.	[SHARE]	ISACs for different industries

			• [SHARE] Relay relevant cross-sector vulnerability & threat intelligence to crypto-focused ISACs and security coalitions	• Share anonymized incident reports, exploit indicators, or vulnerability disclosures that may be relevant to blockchain stakeholders. • Provide briefings or secure bulletins to crypto-specific ISACs on coordinated or large-scale cyber incidents (e.g., DDoS campaigns, supply chain compromises). • Participate in joint intelligence-sharing initiatives or sector-to-sector response exercises with crypto ISAC counterparts. • Enable reciprocal access to threat intelligence platforms or real-time alert systems to strengthen multi-industry situational awareness. • Help blockchain ISACs identify upstream dependencies and systemic risks originating outside the crypto domain.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party Security Expert Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit
			• [CONTRIBUTE TO STANDARDS] Offer domain-specific expertise to improve security and interoperability standards	• Provide feedback on proposed blockchain security standards by sharing analogies or frameworks from their respective sectors. • Contribute insights on incident response coordination, third-party risk governance, and operational resilience frameworks. • Participate in multi-sector standards initiatives (e.g., through NIST, ISO, or BGIN) to help align definitions, metrics, and compliance structures. • Recommend controls and procedures that have successfully reduced threat exposure in their domains and may apply to crypto entities. • Facilitate dialogue between regulators, infrastructure providers, and blockchain developers to build interoperable standards across finance, healthcare, and crypto.	[STANDARDIZE]	International Standards Organization National Standards Organization
	International standards organization	• Technical standards organization (ISO) • Regulatory standards organization (FATF, IOSCO, FSB, etc) • Voluntary-based standards organization for crypto (CCSS, BSSC, etc) • BGIN • OASIS • IETF	• [ANALYZE] Identify global patterns in vulnerabilities and threats to inform standard-setting	• Collect and synthesize vulnerability and incident data from regulators, technical communities, law enforcement, and industry participants. • Analyze the impact of repeated protocol design flaws, smart contract bugs, or governance failures across blockchain systems. • Monitor cross-border laundering methods, financial instability risks, or systemic interdependencies in digital asset ecosystems. • Detect common challenges in implementation, compliance, or enforcement of existing standards, feeding into revision cycles. • Integrate findings from public and private sources to ensure comprehensive, evidence-based standard development.	[SHARE]	International Standards Organization
			• [SHARE] Disseminate threat and risk insights to guide policy, compliance, and technical adoption	• Publish reports, typology analyses, and trend assessments on vulnerabilities, attacks, and system-level weaknesses in blockchain and financial technologies. • Host workshops, roundtables, and public consultations to engage stakeholders in interpreting and acting on emerging threats. • Share anonymized case studies and scenario-based risk modeling to illustrate challenges in threat detection, response, and coordination. • Provide practical guidance, interpretive notes, and risk frameworks to regulators, companies, and developers adopting or aligning with standards. • Issue updates and advisory notes to inform real-time adjustments in security posture and governance practices.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App Consumers/Relying Party Security Expert Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit

			• [STANDARDIZE] Develop internationally recognized frameworks for blockchain security and risk governance	• Facilitate multi-stakeholder working groups to co-create baseline standards for secure system architecture, custody, key management, and smart contract safety. • Translate empirical research and operational lessons into formalized principles, control measures, and testing protocols. • Define taxonomies and reference models (e.g., for vulnerabilities, roles, and assurance levels) to ensure a common language and approach to risk mitigation. • Develop modular, implementation-neutral standards that can be adapted by regulators, enterprises, and open-source communities. • Align blockchain-related standards with broader cybersecurity, privacy, AML/CFT, and financial stability standards used by traditional finance and critical infrastructure.	[PROMOTE STANDARD]	Security Expert Auditor Insurance Company Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authorities Financial Intelligence Units Govt. Labs / Response Teams Crypto-focused ISAC
					[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party
	National Standards Organization	• Technical standards organization (NIST and IPA) • Self-regulatory standards organizations	• [ANALYZE] Analyze trends on vulnerabilities and threats	• Monitor and assess vulnerability disclosures, cybersecurity incident reports, and threat intelligence feeds to identify common weaknesses in blockchain and related systems. • Study incident case studies and exploit methodologies across industries to extract lessons applicable to national infrastructure and regulatory priorities. • Identify systemic risks within the domestic ecosystem, such as protocol dependency on external services, custodial key management failures, or software supply chain issues. • Compare threat trends against existing standards to identify gaps or outdated controls in national frameworks. • Integrate global threat landscape insights with local policy objectives, enforcement realities, and sector-specific needs.	[SHARE]	National Standards Organization
			• [SHARE] Share analysis	• Publish risk assessment reports and technical bulletins on vulnerabilities and attack trends affecting blockchain, financial services, or national critical infrastructure. • Host industry engagement forums and briefings to discuss research findings and promote informed adoption of security practices. • Issue interpretive materials, such as implementation guides and FAQs, to clarify how national actors should apply emerging standards. • Collaborate with government agencies, universities, and security experts to validate analysis and build consensus around national threat response strategies. • Provide sector-specific insights (e.g., for financial services, energy, or digital identity systems) to align blockchain-related risks with existing regulatory efforts.	[ASSESS]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party Security Expert
						Certification Body Law Enforcement Authority Financial Regulatory Authority Financial Intelligence Unit

			• [STANDARDIZE] • Set national security guidelines based on international standards	• Develop and publish national cybersecurity and operational resilience standards informed by global frameworks such as ISO, FATF, NIST SP, or CCSS. • Tailor international standards to the national regulatory environment, economic context, and technological maturity. • Create certification schemes or security baselines for blockchain systems adopted in government, finance, healthcare, or identity management. • Provide normative language for legislation, licensing regimes, or audit procedures referencing the national standards. • Coordinate with sector regulators and ministries to embed these guidelines into supervisory tools, compliance assessments, and innovation frameworks.	[PROMOTE STANDARD]	Security Expert Auditor Insurance Company Blockchain Analytics Firm
						Certification Body Law Enforcement Authority Financial Regulatory Authorities Financial Intelligence Units Govt. Labs / Response Teams Crypto-focused ISAC
					[USE STANDARD]	Blockchain Protocol Blockchain Network Blockchain App
						Consumers/Relying Party